A Lightweight Markdown-to-HTML Paper Converter

CSS-first academic publishing for web and print

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ABSTRACT

This paper describes Paperify, a lightweight Markdown-to-HTML converter designed for high-quality online reading and two-column print output. The converter emits minimal, semantic HTML; a carefully authored stylesheet carries the layout and typography for both media. We show that math, figures, tables, code, footnotes, and video can all be expressed in plain Markdown while remaining printable.

KEYWORDS

Markdown, HTML, CSS, academic publishing

Introduction

Academic writing tools tend to cluster at two extremes: heavyweight LaTeX toolchains with excellent print output, and web-first Markdown renderers with no print story at all. Paperify sits deliberately between them, taking some inspiration from Pandoc's document conversion model [2][1]. The authoring format is plain Markdown, the output is a single portable HTML file, and the *same* document reads comfortably on a phone and prints as a two-column paper.

The design philosophy is simple: keep the HTML semantic and stable, and let the stylesheet do the visual work. When the converter stays out of layout decisions, themes remain possible and documents remain durable.

Method

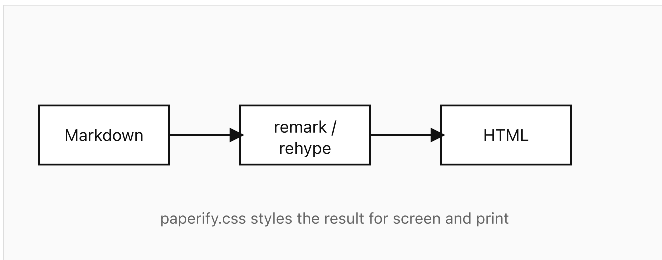
Given a sequence of tokens $x_1, x_2, \ldots, x_n$, we model the joint probability autoregressively:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i \mid x_1, \ldots, x_{i-1})$$

The per-token loss is the negative log-likelihood $\mathcal{L} = -\frac{1}{n}\sum_i \log p(x_i \mid x_{<i})$, rendered here inline to show that math sits naturally within running text.

Architecture overview

A regular Markdown image whose paragraph contains nothing else becomes a semantic figure, with the alt text doubling as the caption:



Column-balanced rendering pipeline of the converter

Wide figures use a directive and span both columns in print:

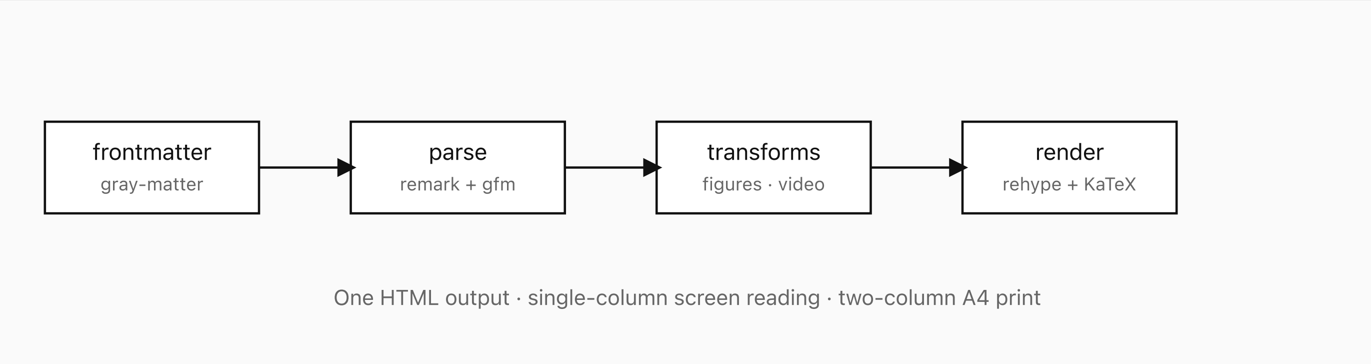
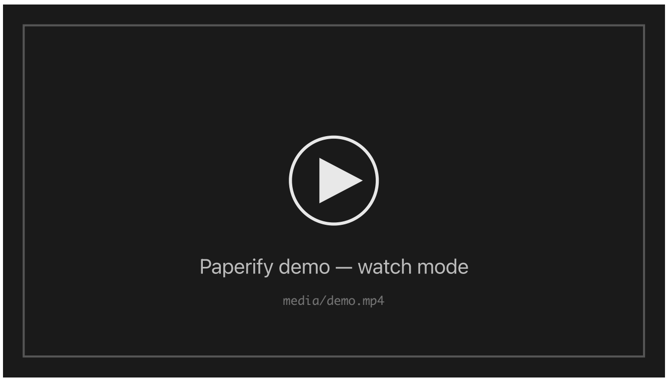


Figure 2: System overview. In print this figure spans the full page width.

Demonstration video

Videos are embedded with a directive. On screen the video is playable; in print the poster frame is shown with a readable source link:



Video available at: media/demo.mp4

A short demonstration of live rebuilds in watch mode.

Evaluation

We compare Paperify against two baselines on document build time and output size. Values are the median of ten runs.

System	Build time (ms)	Output size (KB)	Print layout
Paperify	84	46	two-column
Baseline A	410	212	single
Baseline B	2,930	188	two-column

The conversion pipeline itself is short and builds on unified's structured content model [3]. The core of it looks like this:

```
const processor = unified()
  .use(remarkParse)
  .use(remarkGfm)
  .use(remarkMath)
  .use(remarkDirective)
  .use(paperifyTransforms)
  .use(remarkRehype)
  .use(rehypeKatex)
  .use(rehypeHighlight, { detect: false })
  .use(rehypeStringify);
```

> Design note: every feature in this paper — math, figures, tables, footnotes, video — degrades gracefully. If a directive is removed, the document is still valid Markdown everywhere else.

Discussion

Two-column print layout in browsers is imperfect: column balancing and break control vary by engine[2]. Paperify accepts this trade-off in exchange for a zero-install toolchain. It borrows some typographic discipline from TeX-era publishing without trying to become TeX [1]. Advanced float placement and true page-bottom footnotes are explicitly out of scope for v1.

Conclusion

Paperify shows that a Markdown pipeline plus one disciplined stylesheet is enough for readable, printable academic documents. The converter stays small; the CSS carries the craft.

1. The name follows the convention of verb-ifying nouns, which we neither endorse nor apologize for.
2. Chromium generally produces the most predictable multi-column print output at the time of writing.

References

[1] Donald E. Knuth. 1984. *The TeXbook*. Addison-Wesley.

[2] John MacFarlane. 2006. Pandoc: A Universal Document Converter. Retrieved from https://pandoc.org

[3] The unified collective. 2015. unified: Content as Structured Data. Retrieved from https://unifiedjs.com